Project Report

Machine Learning Method for Study of the Sivers Function

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Certificate

This is to certify that A R Bathri Narayanan, student of UM DAE CEBS, has undertaken project work from 1st January, 2026 to 30th April, 2026 under the guidance of Prof HS Mani, Chennai Mathematical Institute, Chennai and Prof. Anuradha Misra, UM-DAE Centre for Excellence in Basic Sciences, Mumbai.

This submitted report titled Machine Learning Method for Study of the Sivers Function is towards the academic requirements of the Integrated Msc. course at UM DAE CEBS.

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Abstract

In this project, we attempt to study about structure of proton using semi inclusive deep inelastic scattering. We first revise the concepts of deep inelastic scattering and deriving the scattering cross section. Then we extend the idea to polarized deep inelastic scattering, where we consider the spin of both the target and the projectile while calculating the hadron tensor. We extend this idea to semi inclusive deep inelastic scattering, which leads us to new structure functions. We then take one of the functions, and analyze it's properties using classical and quantum machine learning techniques.

Contents

Chapter 1

Introduction

Scattering experiments have played a central role in revealing the internal structure of matter. From the Rutherford gold foil experiment, which established the existence of the atomic nucleus, to modern high-energy collision experiments at particle accelerators such as CERN and Fermilab, scattering processes have continuously provided insight into progressively smaller length scales and fundamental constituents of matter.

One of the most important classes of scattering experiments in modern particle physics is Deep Inelastic Scattering (DIS). In DIS experiments, highly energetic leptons such as electrons or muons are scattered off nucleons. The large momentum transfer involved in the interaction allows probing the internal structure of the proton at very short distances. Unlike elastic scattering, where the target remains intact, deep inelastic scattering causes the proton to break apart into several hadrons, indicating that it possesses an internal partonic structure composed of quarks and gluons.

The development of the parton model and Quantum Chromodynamics (QCD) established that the proton is not an elementary particle, but rather a bound state of strongly interacting quarks and gluons. DIS experiments provided some of the earliest direct evidence for the existence of these constituents and enabled the extraction of Parton Distribution Functions (PDFs), which describe the longitudinal momentum distribution of quarks and gluons inside the proton.

In conventional DIS analyses, the transverse momentum and spin degrees of freedom of the partons are often averaged over. Such an approximation is sufficient for many inclusive processes, where only the scattered lepton is observed. However, a more complete understanding of nucleon structure requires the inclusion of spin and intrinsic transverse momentum effects. Experimental observations have demonstrated that spin-dependent phenomena produce measurable asymmetries that cannot be explained solely using collinear PDFs.

Another channel to study the internal structure of hadrons is provided by Semi-Inclusive Deep Inelastic Scattering (SIDIS), where, in addition to the scattered lepton, one or more hadrons produced in the final state are also detected. SIDIS provides access to the transverse momentum and spin structure of partons inside the nucleon. In particular, it enables the study of Transverse Momentum Dependent Parton Distribution Functions (TMD PDFs), which describe not only the longitudinal momentum fraction carried by a parton, but also its intrinsic transverse momentum and spin correlations.

TMDs provide a three-dimensional momentum-space description of the nucleon and reveal nontrivial correlations between parton motion and nucleon spin. Among these distributions, the Sivers and Boer–Mulders functions are especially important because they encode spin-orbit correlations and generate azimuthal asymmetries observed in polarized scattering experiments.

The study of TMDs therefore plays an essential role in understanding the internal dynamics of hadrons, the orbital motion of partons, and the spin structure of the proton within Quantum Chromodynamics.

1.1 Motivation

Spin is one of the most fundamental quantum mechanical degrees of freedom and experiments with polarized beams plays a crucial role in determining the structure of hadrons. A complete understanding of any quantum field theory requires detailed investigation of spin-dependent scattering processes and decay mechanisms. In hadronic physics, spin observables provide valuable information regarding the internal dynamics and correlations among quarks and gluons inside composite systems such as the proton.

Although the proton has spin $\frac{1}{2}$, understanding how this spin emerges from its constituent quarks and gluons remains one of the major unresolved problems in Quantum Chromodynamics. Experimental studies have shown that the spin carried by quarks alone cannot fully account for the total proton spin, suggesting significant contributions from gluon spin and orbital angular momentum. Consequently, studying spin-dependent observables provides an important pathway toward understanding the internal spin structure of hadrons.

The primary motivation of this project is to use machine learning methods to study a particular type of Transverse Momentum Dependent Parton Distribution Function (TMD) called the Sivers function. TMDs encode correlations between the spin of the nucleon, the spin of the partons, and their intrinsic transverse momentum. These distributions therefore provide a more complete and multidimensional description of proton structure compared to ordinary collinear PDFs. Understanding these distribution is essential for explaining experimentally observed azimuthal asymmetries in Semi-Inclusive Deep Inelastic Scattering processes.

Major motivation behind this work is the application of machine learning techniques to study gluon Sivers function. Modern scattering experiments generate large and highly complex datasets, making machine learning methods increasingly valuable for pattern recognition, classification, and feature extraction. Classical machine learning algorithms provide powerful tools for analyzing multidimensional physical observables and identifying nontrivial structures in the data.

Furthermore, recent developments in quantum computing have led to the emergence of Quantum Machine Learning (QML), which combines quantum information processing with machine learning techniques. Since the underlying physics of hadronic systems is itself quantum mechanical, QML methods may offer new approaches for studying high-dimensional quantum data and spin-dependent observables.

This project aims to explore the applicability of both classical and quantum machine learning methods in the analysis of spin-dependent scattering phenomena and TMD-related observables. By comparing the performance of classical machine learning and quantum machine learning models, the work seeks to evaluate the potential advantages and limitations of quantum-enhanced learning techniques in modern high-energy physics research. The plan of the report is as follows.

1.2 Project outline

Chapter 2 provides an introduction to electron proton scattering, parton model and the derivation of the cross section . Chapter 3 delves deep into deep inelastic scattering. We consider the effects of spin. We obtain the 18 structure functions and transverse momentum distributions (TMDs) . Chapter 4 is a closer look at one of the transverse momentum distributions, the Sivers function. We discuss its connection with the experimentally observed transverse single spin assymmetries. Chapter 5 explains the machine learning aspect of the project, where we use machine learning techniques to analyze a trial problem of segregation of pure and mixed state density matrices. Chapter 6 applies the machine learning models to actual data to obtain the Sivers function. Chapter 7 concludes the project and lists potential future directions one can work on.

Chapter 2

Deep Inelastic Scattering

2.1 Introduction

This section is an introduction to deep inelastic scattering. We start with simple electron proton elastic scattering, where we derive a formula to determine the cross section. Then we introduce inelastic scattering, where we define new variables that describe particle kinematics better in the inelastic regime. Next, we discuss deep inelastic case, where we determine the scattering cross section using new functions called structure functions. Finally, we bring two important features of inelastic scattering (Bjorken scaling and Callan-Gross relation), and introduce parton distribution functions. More details about scattering, including that of matrix element, Rutherford and Mott scattering, can be found in [1][2].

2.2 Electron Proton scattering

2.2.1 Elastic scattering

We shall start with the general case where the system is relativistic and proton has a recoil. Here we consider the proton to be a point particle. The four momenta of the initial and final state are given in proton reference frame as

$$p_1 = (E_1, 0, 0, E_1) \quad p_2 = (m_p, 0, 0, 0), \quad p_3 = (E_3, 0, E_3 \sin\theta, E_3 \cos\theta), \quad p_4 = (E_3, \mathbf{p_4}). \tag{2.1}$$

We can write the matrix element as [2]

$$\langle |\mathcal{M}_{fi}|^2 \rangle = \frac{8e^4}{(p_1 - p_3)^4} [(p_1.p_2)(p_3.p_4) + (p_1.p_4)(p_2.p_3) - m_p^2(p_1.p_3)]. \tag{2.2}$$

It is useful to obtain the matrix element in terms of the energy and scattering angle of the outgoing electron, as

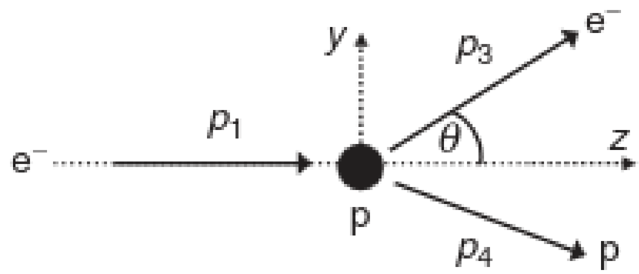

Figure 2.1: Relativistic elastic scattering of an electron by a proton, image taken from [2]

this is the quantity usually measured. p_4 can be eliminated using the conservation of momentum as in equation 2.1. We obtain the other terms as

$$p_2.p_3 = E_3 m_p \quad p_1.p_2 = E_1 m_p \quad p_1.p_3 = E_1 E_3(1 - cos\theta) \tag{2.3}$$

$$p_3.p_4 = p_3.p_2 + p_3.p_2 - p_3.p_3 = E_1E_3(1 - cos\theta) + E_3m_p \tag{2.4}$$

$$p_1.p_4 = p_1.p_2 + p_1.p_2 - p_1.p_3 = -E_1E_3(1 - cos\theta) + E_1m_p \tag{2.5}$$

We have ignored $p_1.p_1 = p_3.p_3 = m_e^2$. The matrix element now becomes

$$\begin{aligned}\langle|\mathcal{M}_{fi}|^2\rangle &= \frac{8e^4}{(p_1 - p_3)^4} m_pE_1E_3 \left[(E_1 - E_3)(1 - cos\theta) + m_p(1 + cos\theta)\right], \\ &= \frac{8e^4}{(p_1 - p_3)^4} 2m_pE_1E_3 \left[(E_1 - E_3)(sin^2\frac{\theta}{2}) + m_p(cos^2\frac{\theta}{2})\right].\end{aligned} \tag{2.6}$$

We can also express the four momentum squared of the virtual photon in terms of E_3 and θ as

$$q^2 = (p_1 - p_3)^2 = p_1^2 + p_3^2 - 2.p_1.p_3 \approx -4E_1E_3sin^2\frac{\theta}{2}, \tag{2.7}$$

where we use the similar argument as before for p_1^2 and p_3^2. It is convenient to work with Q^2 which is defined as

$$Q^2 = -q^2 = 4E_1E_3sin^2\frac{\theta}{2}. \tag{2.8}$$

Energy lost in the scattering process can be expressed in terms of Q^2 by noting that

$$q.p_2 = (p_1 - p_3)p_2 = m_p(E_1 - E_3). \tag{2.9}$$

$q.p_2$ can also be written in terms of $q = p_4 - p_2$ and noting

$$p_4^2 = (q + p_2)^2 = q^2 + 2q.p_2 + p_2^2, \quad p_2^2 = p_4^2 = m_p^2. \tag{2.10}$$

This implies $q.p_2 = -q^2/2$. Therefore the energy difference can be given by

$$E_1 - E_3 = \frac{-q^2}{2m_p} = \frac{Q^2}{2m_p}. \tag{2.11}$$

This equation implies that electron always loses it's initial energy due to scattering. The matrix element can be written as

$$\langle|\mathcal{M}_{fi}|^2\rangle = \frac{m_p^2e^4}{E_1E_3sin^4\theta/2}\left[cos^2\frac{\theta}{2} + \frac{Q^2}{2m_p}sin^2\frac{\theta}{2}\right]. \tag{2.12}$$

We also know that [2]

$$\frac{d\sigma}{d\Omega} \approx \frac{1}{64\pi^2}\left(\frac{E_3}{m_pE_1}\right)^2 \langle|\mathcal{M}_{fi}|^2\rangle \tag{2.13}$$

Hence we obtain the general formula for elastic scattering

$$\frac{d\sigma}{d\Omega} = \frac{\alpha^2}{4E_1sin^4\theta/2}\frac{E_3}{E_1}\left[cos^2\frac{\theta}{2} + \frac{Q^2}{2m_p}sin^2\frac{\theta}{2}\right]. \tag{2.14}$$

Note that both Q^2 and E_3 can be expressed in terms of scattering angle θ.

It can be shown that the most general formula for electron proton elastic scattering in case of an extended proton target is given by the Rosenbleuth formula [2]

$$\frac{d\sigma}{d\Omega} = \frac{\alpha^2}{4E_1^2sin^4(\frac{\theta}{2})}\frac{E_3}{E_1}\left(\frac{G_E^2 + \tau G_M^2}{1 + \tau}cos^2\frac{\theta}{2} + 2\tau G_M^2 sin^2\frac{\theta}{2}\right). \tag{2.15}$$

where $\tau = Q^2/4m_p$ and G_E and G_M are electric and magnetic form factors respectively.

2.2.2 Inelastic scattering

Since the proton has a finite size, the cross section reduces rapidly with an increase in energy. At high energies, inelastic processes dominate, and the proton breaks up into many particles. The invariant mass of the hadron system W, depends on the four momentum of the virtual photon as $W^2 = p_4^2 = (p_2 + q)^2$, and can take a range of values. In elastic scattering, the invariant mass is always the mass of the proton and event kinematics can be described by the electron scattering angle alone. But in the inelastic case, due to this additional degree of freedom, we need two quantities to describe the event kinematics, and we choose one of the Lorentz invariant quantities W, x, y, v and Q^2, which we shall define now.

The definition of $Q^2 = -q^2$ remains the same as that of elastic scattering. We can rewrite Q^2 in terms of electron four momenta as

$$Q^2 = -(p_1 - p_3)^2 = -2m_e^2 + 2p_1p_3 = -2m_e^2 + 2E_1E_3 - 2p_1p_3cos\theta. \tag{2.16}$$

In the inelastic limit, the energy is so large that the mass of the electron can be ignored and hence we obtain

$$Q^2 \approx 2E_1E_3(1 - cos\theta) = 4E_1E_3sin^2\frac{\theta}{2}. \tag{2.17}$$

We shall now introduce Bjorken variable x, which is a Lorentz invariant dimensionless quantity, defined as

$$x = \frac{Q^2}{2p_2.q}. \tag{2.18}$$

To rewrite x in terms of invariant mass, we expand the definition of W

$$\begin{aligned} W^2 = p_4^2 = (q^2 + p_2^2) = q^2 + p_2^2 + 2p_2.q, \\ \implies W^2 + Q^2 - m_p^2 = 2p_2.q. \end{aligned} \tag{2.19}$$

This implies

$$x = \frac{Q^2}{Q^2 + W^2 - m_p^2}. \tag{2.20}$$

Now, as there are three valence quarks in a proton and quark and anti quark are produced only in pairs, there should be at least one baryon in the hadronic final state. This implies the invariant hadronic mass will be always greater than the mass of the proton, which is the lightest baryon. ($W^2 = p_4^2 \geq m_p^2$). That creates a bound on the variable x given by

$$0 \leq x \leq 1. \tag{2.21}$$

In effect, x describes the elasticity of the scattering process. In the extreme case when $x = 1$, the invariant mass is just the mass of the proton, and the process is elastic. We can now define yet another variable y denoting the inelasticity of the process as

$$y = \frac{p_2.q}{p_2.p_1}. \tag{2.22}$$

In the frame where the proton is at rest, $p_2 = (m_p, 0, 0, 0)$ we have

$$y = \frac{m_p(E_1 - E_3)}{m_p.E_1} = 1 - \frac{E_3}{E_1}. \tag{2.23}$$

Hence y is the fractional energy lost by the electon in the frame where the proton is at rest. As the energy of the final state hadronic system is larger than the initial state proton, $E_4 \geq m_p$ which implies the electon should lose energy. Hence y lies between

$$0 \leq y \leq 1. \tag{2.24}$$

Sometimes it is easier to work with energies instead of fractional energies, and hence we define another variable ν, defined as

$$\nu = \frac{p_2.q}{m_p} = E_1 - E_3, \tag{2.25}$$

which is simply the energy lost by the electron. Given the centre of mass energy $\sqrt{s}$, kinematics of the inelastic scattering can be described by two independent observables from any of the Lorentz invariant quantities Q^2, x, y, ν, provided they are independent.

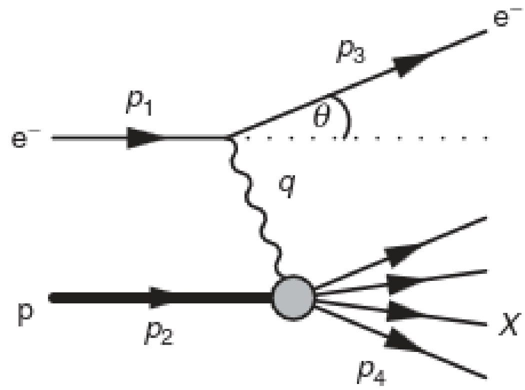

Figure 2.2: Diagrammatic representaion of deep inelastic scattering. Image taken from [2]

2.3 Deep Inelastic Scattering and Scaling

2.3.1 Scattering cross section

We shall start with the most general Lorentz invariant $e^-p \to e^-p$ form given by Rosenbleuth for elastic scattering

$$\frac{d\sigma}{d\Omega} = \frac{\alpha^2}{4E_1^2 sin^4(\frac{\theta}{2})} \frac{E_3}{E_1} \left(\frac{G_E^2 + \tau G_M^2}{1+\tau} cos^2\frac{\theta}{2} + 2\tau G_M^2 sin^2\frac{\theta}{2} \right). \tag{2.26}$$

In terms of the newly defined variables, we obtain

$$\frac{d\sigma}{d\Omega} = \frac{4\pi\alpha^2}{Q^4} \left(\frac{G_E^2 + \tau G_M^2}{1+\tau} \left(1 - y - \frac{m_p^2 y^2}{Q^2} \right) + \frac{y^2}{2} G_M^2 \right). \tag{2.27}$$

The Q^2 dependence of the form factors $G_E(Q^2)$ and $G_M(Q^2)$ and $\tau = Q^2/4m_p$ can be absorbed into two new functions $f_1(Q^2)$ and $f_2(Q^2)$

$$\frac{d\sigma}{d\Omega} = \frac{4\pi\alpha^2}{Q^4} \left(\left(1 - y - \frac{m_p^2 y^2}{Q^2} \right) f_2(Q^2) + \frac{y^2}{2} f_1(Q^2) \right). \tag{2.28}$$

Even though y appears in this equation, this is the case of elastic scattering, so $x = 1$ and $y = f(Q^2)$ alone. Here $f_1(Q^2)$ is associated with purely magnetic interaction and $f_2(Q^2)$ is associated with electric and magnetic interaction. Now this cross section formula can be generalized as

$$\frac{d\sigma}{d\Omega} = \frac{4\pi\alpha^2}{Q^4} \left(\left(1 - y - \frac{m_p^2 y^2}{Q^2} \right) \frac{F_2(x,Q^2)}{x} + y^2 F_1(x,Q^2) \right), \tag{2.29}$$

where the variables $f_1(Q^2)$ and $f_2(Q^2)$ are replaced by $F_1(x,Q^2)$ and $F_2(x,Q^2)$. F_1 is associated with purely magnetic interaction and F_2 is associated with both electric and magnetic field. In deep inelastic limit, $Q^2 \gg m_p^2 y^2$, hence the equation approximates to

$$\frac{d\sigma}{d\Omega} = \frac{4\pi\alpha^2}{Q^4} \left((1-y) \frac{F_2(x,Q^2)}{x} + y^2 F_1(x,Q^2) \right). \tag{2.30}$$

In fixed target experiments, Lorentz invariant variables Q^2, x, y can be obtained by observing energy and scattering angle of e^-, E_3 and θ as

$$Q^2 = 4E_1E_3 sin^2\frac{\theta}{2}, \quad x = \frac{Q^2}{2m_p(E_1 - E_3)}, \quad y = 1 - \frac{E_3}{E_1}, \tag{2.31}$$

where E_1 is the energy of the incident electron.

2.3.2 Bjorken Scaling and Callan-Gross Relations

Experiments done at the Stanford Linear Accelerator Center (SLAC) in California revealed two striking features.

1. Both the structure functions $F_1(x, Q^2)$ and $F_2(x, Q^2)$ are independent of Q^2, i.e.

$$F_1(x, Q^2) \to F_1(x), \quad F_2(x, Q^2) \to F_2(x). \tag{2.32}$$

   The independence of Q^2 suggests that the scattering occurs from a point-like constituent within the proton.

2. The structure functions are related in the deep inelastic case as

$$F_2(x) = 2xF_1(x). \tag{2.33}$$

   This is the Callan-Gross relation. We can explain this by assuming that the inelastic scattering is actually an elastic scattering between the electron and a point like spin half particle within the proton, namely the quarks. In this case, the electric and magnetic contributions are related by the fixed magnetic moment of the Dirac particle. Another noteworthy point is that if the quarks were spin 0, then $F_1(x) = 0$.

2.4 Parton Distribution Functions

The quarks inside the protons interact with each other through the exchange of gluons. The dynamics of this interaction leads to a distribution of quark momenta. These distributions are expressed in terms of Parton Distribution Functions (PDFs). For example, up quark PDF for a proton $u^p(x)$ is defined such that $u^p(x)\delta x$ represents the number of up quarks with a momentum fraction between x and $x + \delta x$. Similarly, one can define a PDF for the down quark, namely $d^p(x)$.

In practice, the functional forms of these PDFs depend on the detailed dynamics of the proton. They are not a priori known and have to be obtained from experiments. The electron proton deep inelastic scattering cross section can be obtained using the PDFs and using the expression for the differential cross section for underlying electron–quark elastic scattering process we had derived. The cross section from a particular flavour of quark i with charge Q_i and momentum fraction in the range x to $x + \delta x$ [2]

$$\frac{d^2\sigma}{dQ^2} = \frac{4\pi\alpha^2}{Q^4}\left[(1-y) + \frac{y^2}{2}\right] Q_i^2 q_i^p(x)\delta x \tag{2.34}$$

We obtain the double differential cross section by dividing the expression by δx and summing over all quark flavours.

$$\implies \frac{d^2\sigma^{exp}}{dxdQ^2} = \frac{4\pi\alpha^2}{Q^4}\left[(1-y) + \frac{y^2}{2}\right] \sum_i Q_i^2 q_i^p(x) \tag{2.35}$$

This is the electron proton deep inelastic scattering cross section in parton model. Comparing with the formula we obtained for cross section using structure function, we deduce.

$$F_2^{exp}(x, Q^2) = 2xF_1^{exp}(x, Q^2) = x\sum_i Q_i^2 q_i^p(x) \tag{2.36}$$

2.5 Conclusion

We have studied scattering of electron and proton in detail, from the elastic case to the deep inelastic case. We also have introduced the notion of structure functions and pointed out some relations which they follow. In the next chapter, we extend the idea further and bring in the contribution of spin and introduce the concept of TMDs

Chapter 3

Transverse Momentum Dependent Distributions (TMDs)

3.1 Introduction

In the last chapter we derived the cross section for deep inelastic scattering. In this chapter, we shall first introduce the hadron tensor and operator definition of PDF. Then we shall introduce the polarized deep inelastic scattering and parametrize the integral correlator introduced in this chapter. Finally, we explain the semi inclusive deep inelastic scattering (SIDIS) and bring out the idea of transverse momentum dependent parton distribution functions (TMD-PDFs) and conclude with TMD factorization.

3.2 The Hadron tensor

In inelastic electron proton scattering, we can write the general expression for the differential cross section as

$$d\sigma \sim L^e_{\mu\nu} W^{\mu\nu} \tag{3.1}$$

Here, $L^e_{\mu\nu}$ represents the lepton tensor

$$L^e_{\mu\nu} = \frac{1}{2} \sum_{e\ spins} [\bar{u}(k')(-e\gamma^\mu)u(k)][\bar{u}(k')(-e\gamma^\mu)u(k)]^* = \frac{1}{2} Tr[(\not{k}' + m)\gamma_\mu(\not{k}' + m)\gamma_\nu] \tag{3.2}$$

The lepton tensor is a symmetric tensor as we have considered unpolarized electrons in the initial states. As we shall see in this chapter, in polarized scattering, we get an antisymmetric contribution also to the lepton tensor. $W^{\mu\nu}$ is the hadron tensor and depends on the independent momenta p and q appearing in the lower vertex of figure 3.1

The general form of the hadron tensor is given by [3].

$$W^{\mu\nu} = W_1(\nu, q^2)\left(-g^{\mu\nu} + \frac{q^\mu q^\nu}{q^2}\right) + W_2(\nu, q^2)\frac{1}{M^2}\left(p^\mu - \frac{p \cdot q}{q^2}q^\mu\right)\left(p^\nu - \frac{p \cdot q}{q^2}q^\nu\right) \tag{3.3}$$

3.3 Operator definition of PDF

In the parton model as discussed in 2, the virtual photon interacts with partons inside the proton incoherently, hence the interaction between the active quark and remnant can be ignored. In this section, we set up a diagrammatic approach for the hadronic processes. We introduce the light cone coordinates

$$a^\mu = (a^+, a^-, \mathbf{a}_\perp), \tag{3.4}$$

where the components in the conventional coordinate system are given as

$$a^\pm = \frac{a^0 \pm a^3}{\sqrt{2}}. \tag{3.5}$$

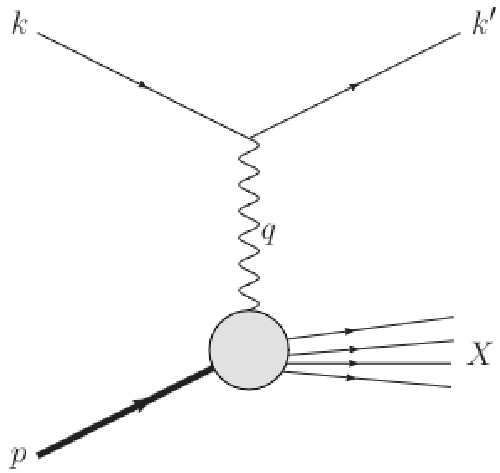

Figure 3.1: Another version of deep inelastic scattering. Figure taken from [3]

The scalar product in terms of the light cone coordinates is given by

$$a.b = a^+b^- + a^-b^+ - \mathbf{a}_\perp.\mathbf{b}_\perp \tag{3.6}$$

We also define light like vectors $n_\pm$

$$a^+ = a.n_-, \tag{3.7a}$$
$$a_- = a.n_+, \tag{3.7b}$$

where, $n_+^2 = n_-^2 = 0$ and $n_+n_- = 1$. We can rewrite P^μ and q^μ as,

$$P^\mu = \left(P^+, \frac{M^2}{2P^+}, \mathbf{0}_\perp\right),$$
$$q^\mu = \left(-x_B P^+, \frac{Q^2}{2x_B P^+}, \mathbf{0}_\perp\right), \tag{3.8}$$

where we take the reference frame with $x_B P^+ = Q/\sqrt{2}$. Consider a final state consisting of a quark with momentum k and a hadronic state X with momentum P_X. Considering the interaction between the electron and quark at the tree level and considering explicit spin dependence of the proton state, the hadron tensor can be written as [3]

$$2MW^{\mu\nu}(q,P,S) = \frac{1}{2\pi}\sum_{e_q} e_q^2 \sum_X \int \frac{d^3P_X}{(2\pi)^3 2P_X^0} \int \frac{d^3k}{(2\pi)^3 2k^0} (2\pi)^4 \delta^4(p+q-k-P_X)$$
$$\times \langle P,S|\, \bar{\psi}_i(0)\, |X\rangle \langle X|\, \bar{\psi}_j(0)\, |P,S\rangle\, \gamma^\mu_{ik}(\not{k}+m)_{kl}\gamma^\nu_{lj}, \tag{3.9}$$

$$= \frac{1}{2\pi}\sum_{e_q} e_q^2 \sum_X \int \delta^4((p+q)^2 - m^2)\Theta(p^0+q^0-m)Tr[\Phi(p,P,S)\gamma^\mu(\not{p}+\not{q}+m)\gamma^\nu], \tag{3.10}$$

where we define the quark-quark correlation function,

$$\Phi_{ij}(p,P,S) = \frac{1}{(2\pi)^4}\int d^4\xi e^{-ip\xi}\langle P,S|\, \bar{\psi}_j(\xi)\psi_i(0)\, |P,S\rangle,$$
$$= \int \frac{d^3P_X}{(2\pi)^3 2P_X^0}\langle P,S|\, \bar{\psi}_j(0)\, |X\rangle \langle X|\, \psi_i(0)\, |P,S\rangle\, \delta^4(P-p-P_X). \tag{3.11}$$

We are not considering flavor for now. The quark-quark correlation function plays an important role in factorization of hadronic processes. As we are interested in a diagrammatic approach for these processes, we would like to know

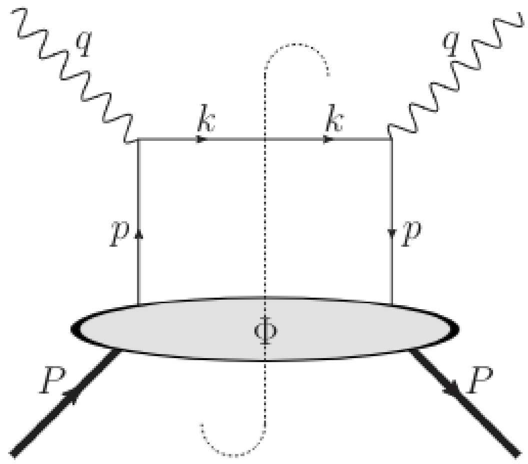

Figure 3.2: The handbag diagram describing hadron tensor $W^{\mu\nu}$ for DIS at tree level in parton model. Image taken from [3]

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about the wavefunctions of the hadrons since while writing amplitude for any process, we need wavefunction for the initial and final states. For point particles, the Fourier expansion of Dirac field in QED is given as

$$\psi(x) = \sum_s \int \frac{d^3k}{(2\pi)^3 2E_k}(u(k,s)e^{-ikx}b(k,s) + v(k,s)e^{ikx}d^\dagger(k,s)), \tag{3.12}$$

which leads to the following expansion for Dirac spinors in terms of matrix element of Dirac field between vacuum and single particle state

$$u(k,s)e^{-ikx} = \langle 0|\,\psi(x)\,|k,s\rangle = \langle 0|\,\psi(x)b^\dagger(k,s)\,|0\rangle\,, \tag{3.13}$$

where the Dirac spinors satisfy the completeness relation

$$\sum_s u(k,s)\bar{u}(k,s) = \not{k} + m \tag{3.14}$$

In the case of scalar fields, the wave functions are just plane wave solutions. In the case of vector fields $A_\mu(x)$ the wave functions are plane wave solutions multiplied by polarization vectors $\epsilon_\mu(k,\lambda)$. Now Lagrangian in QCD is written usually in terms of quarks and gluon fields and not in terms of hadron fields. Quarks and gluons are confined in hadrons and doesn't appear as free particles. Hence we have to rewrite as

$$u(k,s)e^{-ikx} = |P_X\rangle\,\psi(x)\,\langle k,s| = |P_X\rangle\,\psi(0)\,\langle k,s|\,e^{-i(k-P_X)x}, \tag{3.15}$$

such that they follow

$$\sum_s u(k,s)\bar{u}(k,s) = \Phi_{ij}(p,PS) = \frac{1}{(2\pi)^4}\int d^4\xi e^{ip\xi}\,\langle P,S|\,\bar{\psi}_j(\xi)\bar{\psi}_i(0)\,|P,S\rangle\,. \tag{3.16}$$

Usually, when we calculate any observable in QFT, we have to sum all the possible processes that can occur.

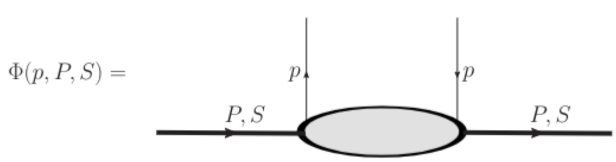


Figure 3.3: A diagrammatic representation of a quark-quark correlator. Image taken from [3]

Since a proton also contains gluons along with quarks, we also have to consider the possibility of a gluon emission contribution. We can simply encode this in the quark-gluon correlation function [3]

$$\Phi^{\mu}_{Aij}(P,P_1;P,S) = \frac{1}{(2\pi)^8}\int d^4\xi \int d^4\eta e^{-i(P-P_1)\xi} e^{-iP_1\eta} \langle P,S| \bar{\psi}_j(\xi) A^{\mu}(\eta)\psi_i(0) |P,S\rangle . \tag{3.17}$$

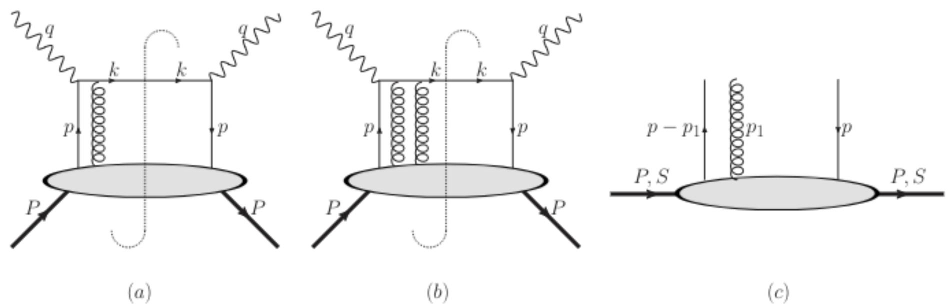


Figure 3.4: The handbag diagram describing hadronic tensor with gluon emission corrections (a) & (b), graphical representation of quark-gluon-quark correlator (c).

We can graphically denote the quark gluon correlation function as shown in figure 3.4. These gluon emissions are needed to ensure the color gauge invariance of the correlation functions.

3.4 Polarized Deep Inelastic Scattering

So far, we have considered the unpolarized deep inelastic scattering cross sections, where we average over the spins of the incoming lepton and proton. The lepton and hadron tensors are symmetric in this case. When we take into account the polarization of the incoming lepton and the target proton, we will get some antisymmetric contribution to the lepton and hadron tensors obtained before. A new class of structure functions can be defined to probe polarized deep inelastic scattering, which can provide vital information about the spin carried by partons. The antisymmetric lepton and hadron tensors are given by [3]

$$L^A_{\mu\nu} = 2m\epsilon_{\mu\nu\rho\sigma} S^{\rho}_l (l-l')^{\sigma}, \tag{3.18}$$

$$W^A_{\mu\nu} = \frac{2M\epsilon_{\mu\nu\rho\sigma}q^{\rho}}{p.q}\left(S^{\sigma}G_1(x,Q^2) + \frac{1}{M}\left(S^{\sigma} - \frac{S.q}{p.q}P^{\sigma}\right)G_2(x,Q^2)\right), \tag{3.19}$$

where s^{μ}_l and S^{μ}_l are the spin 4 vectors of the lepton and hadron respectively. G_1 and G_2 are the structure functions of the hadronic tensor $W^A_{\mu\nu}$. We rewrite $S^{\mu} = S^{\mu}_L + S^{\mu}_{\perp}$, where we essentially split the spin 4 vector into parts which are longitudinal and transverse to the photon axis. The antisymmetric hadron tensor in terms of the new variables can be written as

$$W^A_{\mu\nu} = \frac{2M\epsilon_{\mu\nu\rho\sigma}q^{s}}{p.q}\left(S^{\sigma}_L G_1(x,Q^2) + S^{\sigma}_{\perp}(G_1(x,Q^2)+G_2(x,Q^2))\right). \tag{3.20}$$

Polarized structure function G_1 can be measured in deep inelastic scattering with longitudinally polarized proton and the sum of G_1 and G_2 can be measured with transversely polarized proton. Since at leading twist, $S^{\mu}_L = \mathcal{O}(P^+)$ and $S^{\mu}_{\perp} = \mathcal{O}(1)$. only longitudinal polarization contributes to deep inelastic scattering. Therefore, we obtain

$$W^A_{\mu\nu} = \epsilon_{\mu\nu\rho\sigma}\eta^{\rho}\frac{\sum_a e_a^2}{2} 2MS^{\sigma}_L g_1(x), \tag{3.21}$$

where $g(x)$ is the helicity distribution of the quark that gives information about the spin carried by the parton inside the proton and can be probed in experiments by measuring difference in cross sections with opposite target spin

$$d\sigma^{+s} - d\sigma^{-s} \propto L^A_{\mu\nu}W^{\mu\nu A}. \tag{3.22}$$

Measurement of antisymmetric tensors in polarized deep inelastic scattering provide us the spin structure of proton in terms of its constituents. The measurement of the structure function $G_1(x)$ in 1988 by the European Muon Collaboration at CERN revealed that the sum of the spin of 3 valence electrons contributes very little to the spin of the proton (spin puzzle). Efforts to understand the spin structure of the proton are ongoing.

3.5 The Integral Correlator and Parametrisation

Now, let us focus on the correlation function or correlator defined in equation (3.11). If we integrate over p^- and p_T, we obtain the correlation function that depends only on longitudinal momentum fraction x and is used to define the collinear PDFs,

$$\begin{aligned}\Phi_{ij}(x;P,S) = \Phi(x) &= \int dP^- d^2P_T \Phi(P;P,S) \\ &= \frac{1}{(2\pi)} \int d\xi^- e^{-ip\xi} \langle P,S| \bar{\psi}_j(\xi)\psi_i(0) |P,S\rangle |_{\xi^+=\xi_\perp=0}. \end{aligned} \tag{3.23}$$

This correlator is a matrix in a Dirac space. We have not taken the color index in the above equation, and summation over color is implicit. Since the above correlator contains a product of different fields at different space-time points, it is not a gauge invariant quantity. To make it gauge invariant, a link operator is inserted between the gauge fields.

$$\Phi_{ij}(p,P,S) = \frac{1}{(2\pi)^4} \int d^4\xi e^{-ip\xi} \langle P,S| \bar{\psi}_j(\xi)\mathcal{U}^C(\xi,0)\psi_i(0) |P,S\rangle\,, \tag{3.24}$$

where the link operator or gauge link is given by

$$\mathcal{U}^C(\xi_1,\xi_2) = Pexp[-ig\int_c ds_\mu A^\mu(s)] \tag{3.25}$$

P implies path ordering and c is the integration path connecting the two points ξ_1 and ξ_2. For collinear PDFs, we integrate over transverse momentum of partons in the correlator and hence the separation of fields is from space-time points $[0,\xi^-,0_\perp]$ to $[0,0,0_\perp]$. The gauge field component required to render the gauge invariance of the correlation function is only A^+ which is set to 0 if we work in the light cone gauge in case of collinear PDFs. However, the gauge link will play an important role in the definition of TMD Parton Distribution Functions (TMD-PDFs).

The correlator matrix satisfies the following relations arising from the requirement of hermiticity, parity invariance and time reversal invariance [3].

$$\Phi^\dagger_{ij}(p,P,S) = \gamma^0\Phi_{ij}(p,P,S)\gamma^0, \tag{3.26a}$$

$$\Phi_{ij}(p,P,S) = \gamma^0\Phi_{ij}(\tilde{p},\tilde{P},\tilde{S})\gamma^0, \tag{3.26b}$$

$$\Phi^*_{ij}(p,P,S) = \gamma^5 C\Phi_{ij}(\tilde{p},\tilde{P},\tilde{S})C^\dagger\gamma^5, \tag{3.26c}$$

We write the most general decomposition of $\Phi(x)$ in a basis of Dirac matrices $\{\mathbb{I},\gamma^\mu,\gamma^\mu\gamma^5,i\gamma^5,i\sigma^{\mu\nu}\gamma^5\}$

$$\Phi(p,P,S) = \frac{1}{2}[S\mathbb{I} + \nu_\mu\gamma^\mu + A_\mu\gamma^\mu\gamma^5 + \mathcal{P}_5 i\gamma^5 + \frac{1}{2}\mathcal{T}_{\mu\nu}i\sigma^{\mu\nu}\gamma^5] \tag{3.27}$$

Here, $S,\gamma_\mu,A_\mu,\mathcal{P}_5,\mathcal{T}_{\mu\nu}$ in the above parametrisation are constructed from available vectors of LHS which are p^μ, P^μ and pseudovector S^μ and can be isolated by taking relevant traces with Φ. At leading twist, at order $\mathcal{O}(1/Q)$ the integrated correlator $\Phi(x)$ is parametrized in terms of three leading twist distribution functions

$$\Phi(x) = \frac{1}{2}[f(x)\not{n}_+ + S_L S_L g_1(x)\gamma_5\not{n}_+ + h_1(x)\frac{[\not{S}_+,\not{n}_+]}{2}. \tag{3.28}$$

Here, $f(x), g_1(x)$ and $h_1(x)$ are the unpolarized, helicity and transversity distribution functions. Given by

$$f(x) = \int \frac{d\xi^-}{2\pi} e^{-ip\xi} \langle P,S| \bar{\psi}_j(\xi)\gamma^+\psi_i(0) |P,S\rangle |_{\xi^+=\xi_\perp=0}, \tag{3.29a}$$

$$g_1(x) = \int \frac{d\xi^-}{2\pi} e^{-ip\xi} \langle P,S| \bar{\psi}_j(\xi)\gamma^+\gamma_5\psi_i(0) |P,S\rangle |_{\xi^+=\xi_\perp=0}, \tag{3.29b}$$

$$h_1(x) = \int \frac{d\xi^-}{2\pi} e^{-ip\xi} \langle P,S| \bar{\psi}_j(\xi)\gamma^+\gamma_1\gamma_5\psi_i(0) |P,S\rangle |_{\xi^+=\xi_\perp=0}, \tag{3.29c}$$

We can see that the unpolarized and helicity distributions contribute to unpolarized and longitudinal cross sections. An interesting point is that the transversity distribution, which should have contributed to the cross section of transversely polarized target, is completely absent from DIS process.

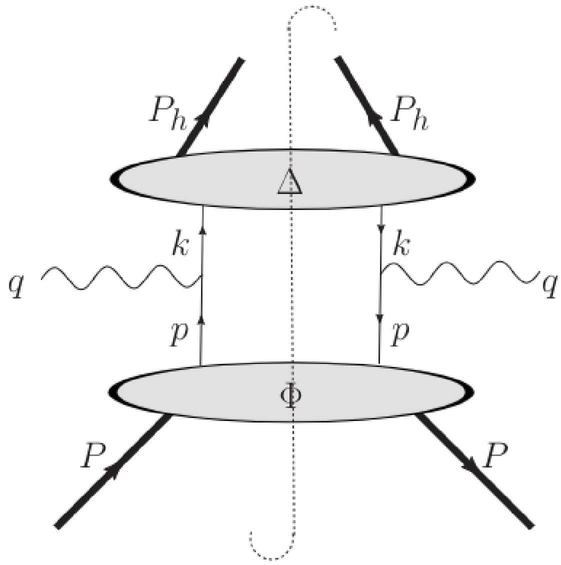


Figure 3.5: The handbag diagram describing hadron tensor for SIDIS at tree level in parton model. Image taken from [3]

3.6 Semi Inclusive Deep Inelastic Scattering

In inclusive deep inelastic scattering, only scattered lepton is observed in the final state. This process enable us to probe PDFs which are functions of only long momentum fraction carried by the parton and the scale. But this doesn't give us information about transverse motion of partons inside proton. Semi Inclusive Deep Inelastic Scattering (SIDIS) provides a way to observe the transverse motion of partons. The cross section for one particle semi inclusive deep inelastic scattering can be written similar to deep inelastic scattering cross section

$$\frac{d^6\sigma}{dx_B.dy.dz_h.d\phi_s.d^2\mathbf{P_{h\perp}}} = \frac{\alpha^2}{4z_h s x_b Q^2} L_{\mu\nu}(l,l',\lambda_e).2MW^{\mu\nu}(q,P,s,P_h) \tag{3.30}$$

where P_h is the momentum of the produced hadron and ϕ_s is the azimuthal angle of the target proton. Along with the variables defined in chapter 2 i.e x_B, y and Q^2, we also define another variable

$$z_h = \frac{P.P_h}{P.l}. \tag{3.31}$$

The hadron tensor for SIDIS now has the P_h dependence, and is given by

$$2MW^{\mu\nu}(q,P,S,P_h) = \frac{1}{(2\pi)^4}\sum_{\chi'}\int\frac{d^3\mathbf{P}_{\chi'}}{2P^0_{\chi'1}} 2\pi\delta^4(q+p-P_{\chi'}-P_h)H^{\mu\nu}(P,S,P_{\chi'},P_h), \tag{3.32}$$

where,

$$H^{\mu\nu}(P,S,P_{\chi'},P_h) = \langle P,S|\, J^\mu(0)\,|P_h,\chi'\rangle\,\langle P_h,\chi'|\, J^\nu(0)\,|P,S\rangle\,, \tag{3.33}$$

where, $P_{\chi'}$ represents the unobserved hadron state. Assuming single photon exchange and considering the final hadron state to be spinless or the spin is not measured, we follow the same procedure as the DIS calculations, the hadron tensor can be written in terms of quark-quark correlators Φ and Δ where Φ is the correlator corresponding to PDFs and Δ is the correlator corresponding to fragmentation of final quark into a hadron

$$2MW^{\mu\nu}(q,P,S,P_h) = \sum_{e_q} e_q^2 \int d^4p.d^4k\delta^4(p+q-k)Tr[\Phi(p,P,S)\gamma^\mu\Delta(k,P_h)\gamma^\nu]. \tag{3.34}$$

The expression of the Δ correlator is given by

$$\begin{aligned}\Delta_{ki}(k,P_h) &= \frac{1}{(2\pi)^4}\int d\xi e^{-ik\xi}\,\langle P,S|\,\psi_k(0)\,|P_h\rangle\,\langle P_h|\,\bar{\psi}_i(\xi)\,|0\rangle \\ &= \sum_Y \int \frac{d\mathbf{P}_Y}{(2\pi)^3 2P_Y^0}\,\langle P,S|\,\psi_k(0)\,|P_h,Y\rangle\,\langle P_h,Y|\,\bar{\psi}_i(\xi)\,|0\rangle\,\delta^4(k-P_h-P_Y)\end{aligned} \tag{3.35}$$

Considering the explicit difference of the beam and target polarization, the hadron tensor is parametrized into 18 structure functions [3][4]

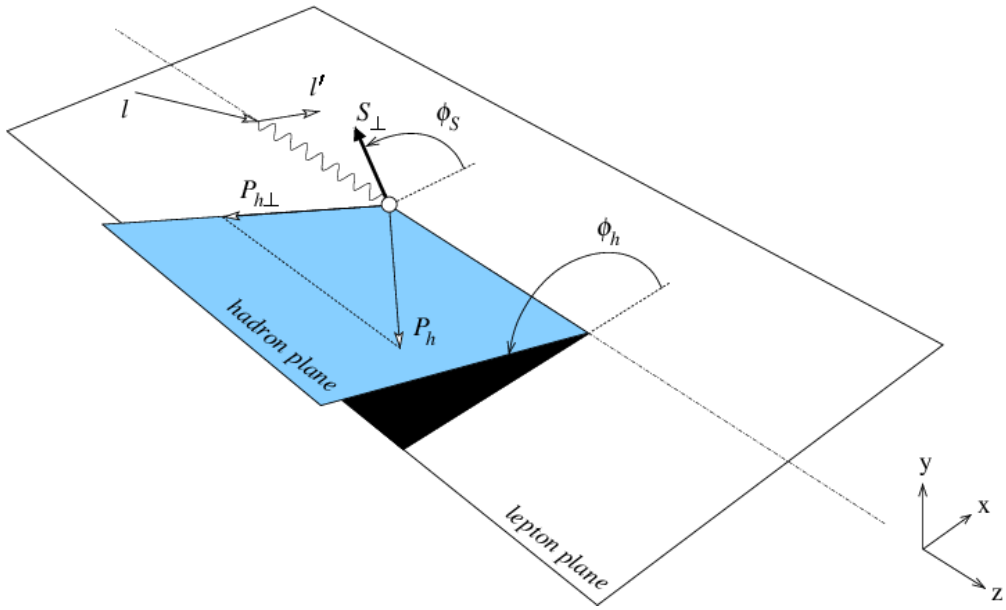


Figure 3.6: SIDIS process and important quantities. Image taken from [4]

$$\begin{aligned}\frac{d^6\sigma}{dx\,dy\,d\phi_S\,dz\,d\phi_h\,dP_{h\perp}^2} =& \frac{\alpha^2}{xyQ^2}\frac{y^2}{2(1-\varepsilon)}\left(1+\frac{2rM^2}{Q^2}\right) \\ &\Big\{F_{UU,T}+\varepsilon\,F_{UU,L}+\sqrt{2\varepsilon(1+\varepsilon)}\cos\phi_h\,F_{UU}^{\cos\phi_h} \\ &+\varepsilon\cos(2\phi_h)\,F_{UU}^{\cos 2\phi_h}+\lambda_e\sqrt{2\varepsilon(1-\varepsilon)}\sin\phi_h\,F_{LU}^{\sin\phi_h} \\ &+S_\parallel\left[\sqrt{2\varepsilon(1+\varepsilon)}\sin\phi_h\,F_{UL}^{\sin\phi_h}+\varepsilon\sin(2\phi_h)\,F_{UL}^{\sin 2\phi_h}\right] \\ &+S_\parallel\lambda_e\left[\sqrt{1-\varepsilon^2}\,F_{LL}+\sqrt{2\varepsilon(1-\varepsilon)}\cos\phi_h\,F_{LL}^{\cos\phi_h}\right] \\ &+|\mathbf{S}_\perp|\Big[\sin(\phi_h-\phi_S)\left(F_{UT,T}^{\sin(\phi_h-\phi_S)}+\varepsilon\,F_{UT,L}^{\sin(\phi_h-\phi_S)}\right) \\ &+\varepsilon\sin(\phi_h+\phi_S)\,F_{UT}^{\sin(\phi_h+\phi_S)}+\varepsilon\sin(3\phi_h-\phi_S)\,F_{UT}^{\sin(3\phi_h-\phi_S)} \\ &+\sqrt{2\varepsilon(1+\varepsilon)}\sin\phi_S\,F_{UT}^{\sin\phi_S}+\sqrt{2\varepsilon(1+\varepsilon)}\sin(2\phi_h-\phi_S)\,F_{UT}^{\sin(2\phi_h-\phi_S)}\Big] \\ &+|\mathbf{S}_\perp|\lambda_e\Big[\sqrt{1-\varepsilon^2}\cos(\phi_h-\phi_S)\,F_{LT}^{\cos(\phi_h-\phi_S)}+\sqrt{2\varepsilon(1-\varepsilon)}\cos\phi_S\,F_{LT}^{\cos\phi_S} \\ &+\sqrt{2\varepsilon(1-\varepsilon)}\cos(2\phi_h-\phi_S)\,F_{LT}^{\cos(2\phi_h-\phi_S)}\Big]\Big\}\end{aligned} \tag{3.36}$$

where the functions depend on the variables x, Q^2, z_h and $P_{h\perp}^2$. The first two subscripts indicate the polarization of the beam and target and the third subscript indicates that of the virtual photon polarization. ϕ_h is the azimuthal angle of the produced hadron in the target rest frame. ϵ is the ration of longitudinal and transverse photon flux, given by

$$\epsilon = \frac{1-y-\frac{1}{4}\gamma^2y^2}{1-y+\frac{1}{2}y^2+\frac{1}{4}\gamma^2y^2}\quad \gamma=\frac{2Mx_B}{Q} \tag{3.37}$$

Integration of 3.36 over transverse momentum of the hadron $P_{h\perp}$ gives the SIDIS cross section

$$\frac{d\sigma}{dx_B\, dy\, d\phi_S\, dz} = \frac{2\alpha^2}{x_B y Q^2}\Bigg\{F_{UU,T} + \varepsilon\, F_{UU,L} + S_{\parallel}\lambda_e\sqrt{1-\varepsilon^2}\, F_{LL} \tag{3.38}$$

$$+|\mathbf{S}_{\perp}|\sqrt{2\varepsilon(1+\varepsilon)}\sin\phi_S\, F_{UT}^{\sin\phi_S} + |\mathbf{S}_{\perp}|\lambda_e\sqrt{2\varepsilon(1-\varepsilon)}\cos\phi_S\, F_{LT}^{\cos\phi_S}\Bigg\} \tag{3.39}$$

where the structure functions in Eq. 2.37 can be obtained from those of Eq. 2.35 by integrating over $P_{h\perp}$, i.e.

$$F_{UU,T}(x_B,\, z_h,\, Q^2) = \int d^2\mathbf{P}_{h\perp}\, F_{UU,T}(x_B,\, z_h,\, P_{h\perp}^2,\, Q^2) \tag{3.40}$$

One can relate the cross section of SIDIS with that of inclusive DIS by integrating the SIDIS cross section over z_h variable and summing over all the hadrons in the final state

$$\frac{d\sigma^{lp\to lX}}{dx_B\, dy\, d\phi_S} = \sum_h \int dz_h\, z_h \frac{d\sigma^{lp\to lhX}}{dz_h\, dx_B\, dy\, d\phi_S\, dz} \tag{3.41}$$

We can identify the following relations between the structure functions of SIDIS and inclusive DIS

$$\sum_h \int dz_h\ z_h\, F_{UU,T}(x_B, z_h, Q^2) = 2x_B F_1(x_B, Q^2) \tag{3.42}$$

$$\sum_h \int dz_h\ z_h\, F_{UU,L}(x_B, z_h, Q^2) = (1+\gamma^2)F_2(x_B, Q^2) - 2x_B F_1(x_B, Q^2) \tag{3.43}$$

$$\sum_h \int dz_h\ z_h\, F_{LL}(x_B, z_h, Q^2) = 2x_B\big(G_1(x_B, Q^2) - \gamma^2 G_2(x_B, Q^2)\big) \tag{3.44}$$

$$\sum_h \int dz_h\ z_h\, F_{LT}^{\cos\phi_S}(x_B, z_h, Q^2) = -2x_B\gamma\big(G_1(x_B, Q^2) + G_2(x_B, Q^2)\big) \tag{3.45}$$

3.7 Intrinsic transverse motion of partons

For SIDIS, in the region where the transverse momentum of the hadron is lower than Q, it has been shown that the cross section can be expressed in terms of transverse-momentum-dependent parton distribution functions (TMD-PDFs) and fragmentation functions (TMD-FFs), collectively referred to as TMDs, i.e

$$\frac{d\sigma^{lp\to lhX}}{dx_B dQ^2 dz_h d^2\mathbf{P}_T} = \sum_q \int d^2k_{\perp} f_q(x, k_{\perp})\frac{d\sigma^{lq\to lq}}{dQ^2} J\frac{z}{z_h} D_q^h(z, p_{\perp}) \tag{3.46}$$

where,

$$x = \frac{1}{2}x_B\left(1+\sqrt{1+\frac{4k_{\perp}^2}{Q^2}}\right) \tag{3.47}$$

the function J is given by,

$$J = \frac{x_B}{x}\left(1+\frac{x_B^2}{x^2}\frac{k_{\perp}^2}{Q^2}\right)^{-1} \tag{3.48}$$

$D_q^h(z, p_{\perp})$ is the TMD fragmentation function describing the fragmentation of parton q into hadron h and $f_q(x, k_{\perp})$ is the transverse momentum dependent PDF. The above factorization of hadronic cross section in terms of unintegrated PDFs and FFs is known as TMD factorization. The transverse motion of partons leads to many interesting effects in scattering, which cannot be explained by standard collinear factorization at leading order.

3.8 Conclusion

In this chapter we have introduced the idea of transverse momentum dependent distribution functions. We first started with the hadron tensor and the operator definition of PDFs as a continuation of the previous chapter. Then we introduced the idea of polarized deep inelastic scattering and parametrised the integral correlator. We then went through the concept of semi inclusive deep inelastic scattering and brought out the 18 structure functions. Finally we introduced the notion of the transverse-momentum-dependent functions and explained the idea of TMD factorization. In the next chapter, we look at one of the TMDs which is of great interest today, that is the Sivers function.

Chapter 4

Sivers function

4.1 Introduction

In this section, we shall look at the Sivers function, it's origin and it's properties. Firstly, we shall obtain the Sivers function from the correlator parametrisation. Then we shall look deeply into gluon TMDs and finally we end this chapter with gauge invariance and process dependence of Sivers function, and have a brief introduction to transverse single spin assymmetries. The chapter follows [3] and [4].

4.2 Sivers Function

TMD-PDFs are derived from the following unintegrated version of correlator of 3.11 i.e.

$$\begin{aligned}\Phi(x,p_T;P,S) \equiv \Phi(x,p_T) &= \int dk^- \Phi(p;P,S) \\ &= \int \frac{d\xi^- d^2\xi_T}{(2\pi)^3} e^{ip\cdot\xi} \left.\langle P,S|\bar{\psi}_j(0)\mathcal{U}^C(0,\xi)\psi_i(\xi)|P,S\rangle\right|_{\xi^+=0}\end{aligned} \tag{2.50}$$

We will refer to the above correlator as TMD correlator. While parameterizing the TMD correlator, we have an extra momentum available which is p_T and more terms appear in the distribution as compared to unintegrated correlator. In the leading-twist approximation, we have [3]

$$\begin{aligned}\Phi(x,p_T) = f_1 \not{n}_+ &- \frac{\epsilon_T^{\alpha\beta} p_{T\alpha} S_{T\beta}}{M} f_{1T}^{\perp} \not{n}_+ \\ &+ S_L g_{1L}\gamma_5 \not{n}_+ - \frac{p_T\cdot S_T}{M} g_{1T}\gamma_5 \not{n}_+ \\ &+ h_{1T}\frac{[\not{S}_T,\not{n}_+]}{2}\gamma_5 + h_{1s}^{\perp}\frac{[\not{p}_T,\not{n}_+]}{2M}\gamma_5 + ih_1^{\perp}\frac{[\not{p}_T,\not{n}_+]}{2M}\end{aligned} \tag{4.1}$$

The distributions f, g and h appearing in above equation are unpolarized, longitudinally polarized and transversely polarized quarks respectively. L (T) refers to the spin of the target being along the longitudinal (transverse) direction; the "$\perp$" symbols signal an explicit dependence on transverse momenta with an uncontracted index. $h_1^{\perp}$ is called Boer-Mulders function and $f_{1T}^{\perp}$ is called Sivers function. These two distributions are T-odd distributions, i.e. they change sign under naive time reversal, which is defined as usual time reversal but without interchange of initial and final states. We have summarized eight independent leading twist quark TMDs in figure 4.1.

A similar thing can be done for TMD FFs, the summary of which is given in the figure 4.2.

Considering unpolarized quarks and transverse spin polarization of proton, we can write

$$\begin{aligned}f_{q/p^\uparrow} &= f_1(x,p_T) - \frac{\epsilon_T^{\alpha\beta}\, p_{T\alpha}\, S_{T\beta}}{M}\, f_{1T}^{\perp}(x,p_T), \\ &= f_1(x,p_T) - \frac{(\hat{\mathbf{P}}\times\mathbf{p}_T)\cdot\mathbf{S}}{M}\, f_{1T}^{\perp}(x,p_T).\end{aligned} \tag{4.2}$$

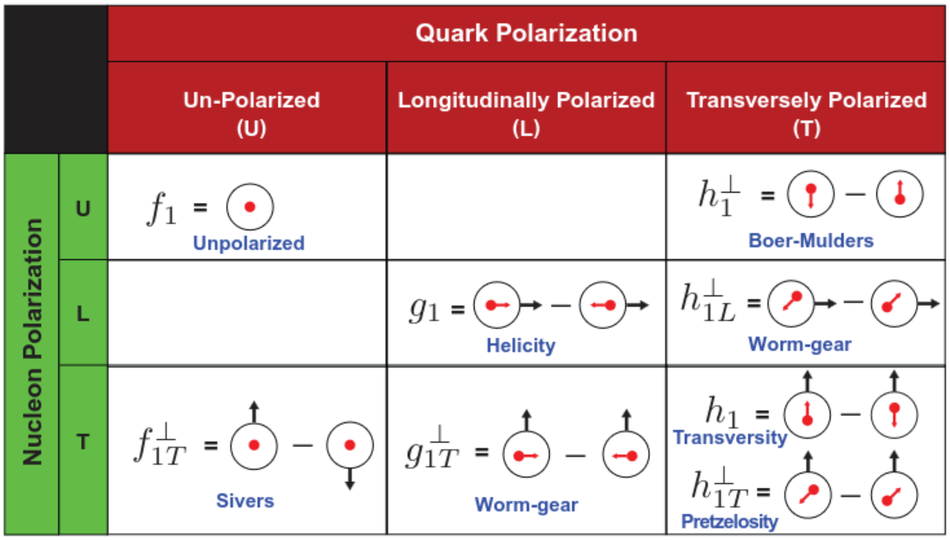


Figure 4.1: The eight independent leading twist quark TMD PDFs, Image taken from [4]

The second part of equation 4.2 holds true in any frame where n and the direction of the proton momentum point in opposite directions. $f_{1T}^{\perp}(x, p_T) > 0$ corresponds to a preference of the quark to move to the left if the proton is moving towards the observer and the proton spin is pointing upwards. Therefore, Sivers distribution measures the anisotropy of the distribution of quark in transversely polarized proton. This asymmetry of distribution of partons in turn give rise to asymmetry in the distribution of produced hadrons in polarized experiments. In the literature, another notation, $\Delta^N f(x, p_T)$ is used for Sivers function which is related to $f_{1T}^{\perp}(x, p_T)$ by [3]

$$\Delta^N f(x, p_T) = -2\,\frac{p_T}{M}\, f_{1T}^{\perp}(x, p_T) \tag{4.3}$$

4.3 Gluon TMDs

The gluon distributions can be calculated using the following gluon-gluon correlator,

$$\Gamma^{\mu\nu;\sigma\rho}(k; P, S) = \int \frac{d^4\xi}{(2\pi)^4} e^{ik\cdot\xi} \langle P, S| F^{\mu\nu}(0)\mathcal{U}(0,\xi) F^{\sigma\rho}|P, S\rangle, \tag{4.4}$$

where, $F_{\mu\nu}(\xi) \equiv F^a_{\mu\nu}(\xi)T^a$ is the field tensor, related to the gauge potential by $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu - ig[A_\mu, A_\nu]$.

It was shown using simple power counting that the most important contributions from $\Gamma^{\mu\nu;\sigma\rho}$ are the ones with the largest possible number of + indices. Due to the antisymmetric character of the field tensor, these are obviously $\Gamma^{+i;+j}$ referred to as twist two, followed by $\Gamma^{+i;+-}$ and $\Gamma^{ij;l+}$ referred to as twist three, where i, j, l, ... indicate transverse indices.

Thus, the correlator for the leading twist gluon TMDs becomes

$$\Gamma^{+i;+j} \equiv \Gamma^{ij}(x, k_\perp) = \int \frac{d\xi^- d^2\xi_T}{(2\pi)^3}\; e^{ip\cdot\xi} \langle P, S| F^{+j}(0)\mathcal{U}(0,\xi) F^{+i}|P, S\rangle\big|_{\xi^+=0} \tag{4.5}$$

Following constraints from Hermiticity, parity conservation and Time reversal invariance, the general decomposition of the correlator in is given in terms of 8 twist two gluon TMDs [3],

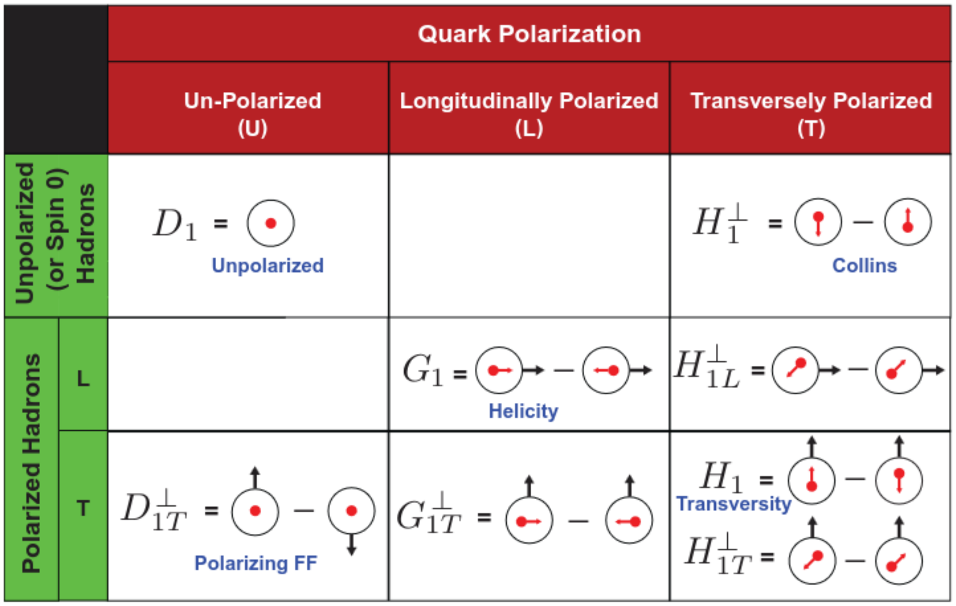


Figure 4.2: The eight independent leading twist quark TMD FFs. Image taken from [4]

$$\begin{aligned}
\Gamma(x, p_T) &= \delta_T^{ij}\Gamma^{ij}(x, p_T) \\
&= f_1^g - \frac{\epsilon_T^{\alpha\beta} p_{T\alpha} S_{T\beta}}{M} f_{1T}^{\perp g} && (4.6) \\
\tilde{\Gamma}(x, p_T) &= i\epsilon_T^{ij}\Gamma^{ij}(x, p_T) \\
&= S_L g_{1L}^g \gamma_5 \not{n}_+ - \frac{p_T \cdot S_T}{M} g_{1T}^g \gamma_5 \not{n}_+ && (4.7) \\
\Gamma_T(x, p_T) &= -\hat{\mathbf{S}}\Gamma^{ij}(x, p_T) \\
&= -\frac{\hat{\mathbf{S}} p_T^i p_T^j}{2M^2} h_1^{\perp g}(x, p_T) + \frac{S_L \hat{\mathbf{S}} p_T^i \epsilon_T^{jk} p_T^k}{2M^2} h_{1L}^{\perp g} \\
&+ \frac{\hat{\mathbf{S}} p_T^i \epsilon_T^{jk} p_T^k}{2M}\left(h_{1T}^g(x, p_T) + \frac{p_T^2}{2M^2} h_{1T}^{\perp g}(x, p_T)\right) \\
&+ \frac{\hat{\mathbf{S}} p_T^i \epsilon_T^{jk}(2p_T^k p_T \cdot S_T - S_T^k p_T^2)}{4M^3} h_{1T}^{\perp g}(x, p_T) && (4.8)
\end{aligned}$$

4.4 Transverse single spin asymmetries

In order to extract the Sivers function, we measure an observable called transverse single spin asymmetry (TSSA), which is defined for the inclusive process $A^\uparrow B \to CX$ as,

$$A_N \equiv \frac{d\sigma^\uparrow - d\sigma^\downarrow}{d\sigma^\uparrow + d\sigma^\downarrow} \tag{4.9}$$

where $d\sigma^{\uparrow(\downarrow)}$ is the cross section when one of the initial nucleons is polarized along the transverse direction $\uparrow$ ($\downarrow$) with respect to the production plane. Within the framework of TMD factorization, A_N can be written as [3],

$$\begin{aligned}
A_N &= \frac{\sum_{a,b,c,d}\int dx_a d^2\mathbf{k}_{\perp a}\, dx_b d^2\mathbf{k}_{\perp b} dz\, d^2\mathbf{k}_{\perp C}\left[f_{a/p^\uparrow}(x_a, k_{\perp a}) - f_{a/p^\downarrow}(x_a, k_{\perp a})\right] f_{b/p}(x_b, k_{\perp b}) H_U^{ab\to cd} D_{C/c}(z, k_{\perp C})}{\sum_{a,b,c,d}\int dx_a d^2\mathbf{k}_{\perp a}\, dx_b d^2\mathbf{k}_{\perp b} dz\, d^2\mathbf{k}_{\perp C}\left[f_{a/p^\uparrow}(x_a, k_{\perp a}) + f_{a/p^\downarrow}(x_a, k_{\perp a})\right] f_{b/p}(x_b, k_{\perp b}) H_U^{ab\to cd} D_{C/c}(z, k_{\perp C})} \\
&= \frac{\sum_{a,b,c,d}\int dx_a d^2\mathbf{k}_{\perp a}\, dx_b d^2\mathbf{k}_{\perp b} dz\, d^2\mathbf{k}_{\perp C} \Delta^N f(x_a, k_{\perp a}) f_{b/p}(x_b, k_{\perp b}) H_U^{ab\to cd} D_{C/c}(z, k_{\perp C})}{\sum_{a,b,c,d}\int dx_a d^2\mathbf{k}_{\perp a}\, dx_b d^2\mathbf{k}_{\perp b} dz\, d^2\mathbf{k}_{\perp C} 2 f_{a/p}(x_a, k_{\perp a}) f_{b/p}(x_b, k_{\perp b}) H_U^{ab\to cd} D_{C/c}(z, k_{\perp C})} && (4.10)
\end{aligned}$$

where, $H_U^{ab\to cd}$ are the unpolarized cross sections of the partonic processes involved. Thus, the numerator of TSSA i.e. equation 4.9 is proportional to the Sivers function $\Delta^N f(x, k_\perp)$ of the initial nucleon A. Since Sivers function has to satisfy the positivity bound,

$$\frac{|\Delta^N f(x, k_\perp)|}{2f(x, k_\perp)} \leq 1 \quad \forall x, k_\perp \tag{4.11}$$

the value of TSSA always lie between -1 and 1.

TSSA, or analyzing power, A_N , is sometimes also referred as "left-right" asymmetry since, by rotational invariance, the number of left going hadrons C when A is downwards polarized, $d\sigma^\downarrow$, is the same as the number of right going hadrons C with A upwards polarized. Here, left (right) is w.r.t. the plane spanned by the momentum and spin directions of the polarized hadron A along its direction of motion.

4.5 Conclusion

This chapter introduced the Sivers function, and it's properties. Firstly, we obtained the Sivers function from the correlator parametrisation, then studied gluon TMDs and finally gave a brief introduction to transverse single spin assymmetries. In the next chapter, we shall look at application of machine learning models in finding the Sivers function.

Chapter 5

Introduction to Machine Learning Models

5.1 Introduction

Machine learning (ML) is a branch of artificial intelligence that develops algorithms capable of learning patterns and relationships from data. Instead of relying on explicitly programmed rules, machine learning models infer functional mappings through statistical optimization procedures.

Broadly, machine learning models attempt to learn a mapping

$$f : \mathbf{x} \to y, \tag{5.1}$$

where $\mathbf{x}$ denotes an input feature vector and y represents the target output.

Machine learning methods may be broadly classified into classical machine learning and quantum machine learning (QML). Classical methods operate on conventional computational architectures, whereas QML employs quantum circuits and quantum-state evolution to construct learning models.

In this work, machine learning techniques were applied to:

1. classification of quantum density matrices into pure and mixed states,

2. extraction of the Sivers function from experimental data of M. Boglione *et al.* [5].

We shall look at the second point in much more detail in chapter 6. A brief overview of the machine learning techniques employed in this work is presented below.

5.2 Classical Machine Learning

Classical machine learning algorithms operate on numerical feature vectors and learn patterns through optimization of statistical loss functions. The input data is represented as

$$\mathbf{x} = (x_1, x_2, \ldots, x_n), \tag{5.2}$$

where each component corresponds to a measurable feature.

Prior to training, feature vectors were standardized according to

$$x' = \frac{x - \mu}{\sigma}, \tag{5.3}$$

where μ and σ denote the mean and standard deviation respectively.

Several supervised learning algorithms were employed in this work.

5.2.1 Support Vector Machines

Support Vector Machines (SVMs) construct separating hyperplanes in feature space:

$$\mathbf{w}^T\mathbf{x} + b = 0. \tag{5.4}$$

For nonlinear classification, Radial Basis Function (RBF) kernels were employed:

$$K(x_i, x_j) = \exp\left(-\gamma||x_i - x_j||^2\right). \tag{5.5}$$

Kernel methods implicitly map feature vectors into higher-dimensional spaces, allowing nonlinear decision boundaries to be constructed.

5.2.2 Logistic Regression and Ensemble Methods

Logistic regression models class probabilities through the sigmoid function:

$$P(y = 1|x) = \frac{1}{1 + e^{-(\mathbf{w}^T\mathbf{x}+b)}}. \tag{5.6}$$

Ensemble methods such as Random Forests and Gradient Boosting combine multiple weak learners to improve predictive accuracy and reduce overfitting.

5.2.3 Artificial Neural Networks

Artificial Neural Networks (ANNs) consist of interconnected layers of neurons performing nonlinear transformations:

$$h = W\mathbf{x} + b. \tag{5.7}$$

The Rectified Linear Unit (ReLU) activation function was employed:

$$\mathrm{ReLU}(x) = \max(0, x). \tag{5.8}$$

The network parameters were optimized using backpropagation and gradient descent.

5.3 Quantum Machine Learning

Quantum machine learning combines quantum computation with machine learning algorithms by encoding classical information into quantum states evolving in Hilbert space.

A general QML model may be written as

$$f_\theta(x) = \langle\psi(x,\theta)|M|\psi(x,\theta)\rangle, \tag{5.9}$$

where $|\psi(x,\theta)\rangle$ is a parameterized quantum state and M is a measurement observable.

5.3.1 Quantum Feature Encoding

Classical data was embedded into quantum states through feature maps:

$$x \to |\phi(x)\rangle. \tag{5.10}$$

The encoding was implemented using rotational quantum gates such as

$$R_X(\theta), \quad R_Y(\theta), \quad R_Z(\theta). \tag{5.11}$$

5.3.2 Quantum Kernel Methods

Quantum kernel methods compute similarities using quantum-state overlaps:

$$K(x_i, x_j) = |\langle\phi(x_i)|\phi(x_j)\rangle|^2. \tag{5.12}$$

The resulting kernel matrix was supplied to a classical SVM classifier.

5.3.3 Variational Quantum Circuits

Variational quantum classifiers employ parameterized quantum circuits of the form

$$|\psi(x,\theta)\rangle = U(\theta)E(x)|0\rangle, \tag{5.13}$$

where $E(x)$ is the encoding circuit and $U(\theta)$ is a trainable variational ansatz.

The model parameters were optimized through hybrid quantum-classical optimization loops. Gradients were evaluated using the parameter-shift rule:

$$\frac{\partial f}{\partial \theta} = \frac{f(\theta + \pi/2) - f(\theta - \pi/2)}{2}. \tag{5.14}$$

5.4 Machine Learning Classification of Density Matrices

Machine learning and quantum machine learning models were employed to classify quantum density matrices into pure and mixed states. This trial project was motivated by [6].

The workflow consisted of:

1. generation of density matrices,
2. extraction of physical observables,
3. preprocessing and feature scaling,
4. training of classical and quantum models,
5. evaluation of classification performance.

5.4.1 Generation of Density Matrices

A valid density matrix ρ satisfies:

$$\rho^\dagger = \rho, \tag{5.15}$$

$$\mathrm{Tr}(\rho) = 1, \tag{5.16}$$

and

$$\rho \geq 0. \tag{5.17}$$

Pure states satisfy

$$\rho^2 = \rho, \tag{5.18}$$

with purity

$$\mathrm{Tr}(\rho^2) = 1, \tag{5.19}$$

whereas mixed states satisfy

$$\mathrm{Tr}(\rho^2) < 1. \tag{5.20}$$

5.4.2 Feature Extraction

Several physically motivated observables were extracted from the density matrices:

- purity $\mathrm{Tr}(\rho^2)$,
- higher-order purity moment $\mathrm{Tr}(\rho^3)$,
- von Neumann entropy,
- eigenvalue statistics,
- Frobenius norm,
- spectral gap,
- rank estimation.

The features were standardized prior to training.

5.4.3 Classical Machine Learning Models

Classical machine learning models including SVMs, Random Forests, Gradient Boosting, Logistic Regression, and Multi-Layer Perceptrons were trained on the extracted feature vectors.

The best-performing classical models achieved nearly perfect classification accuracy when multiple physically motivated observables were included.

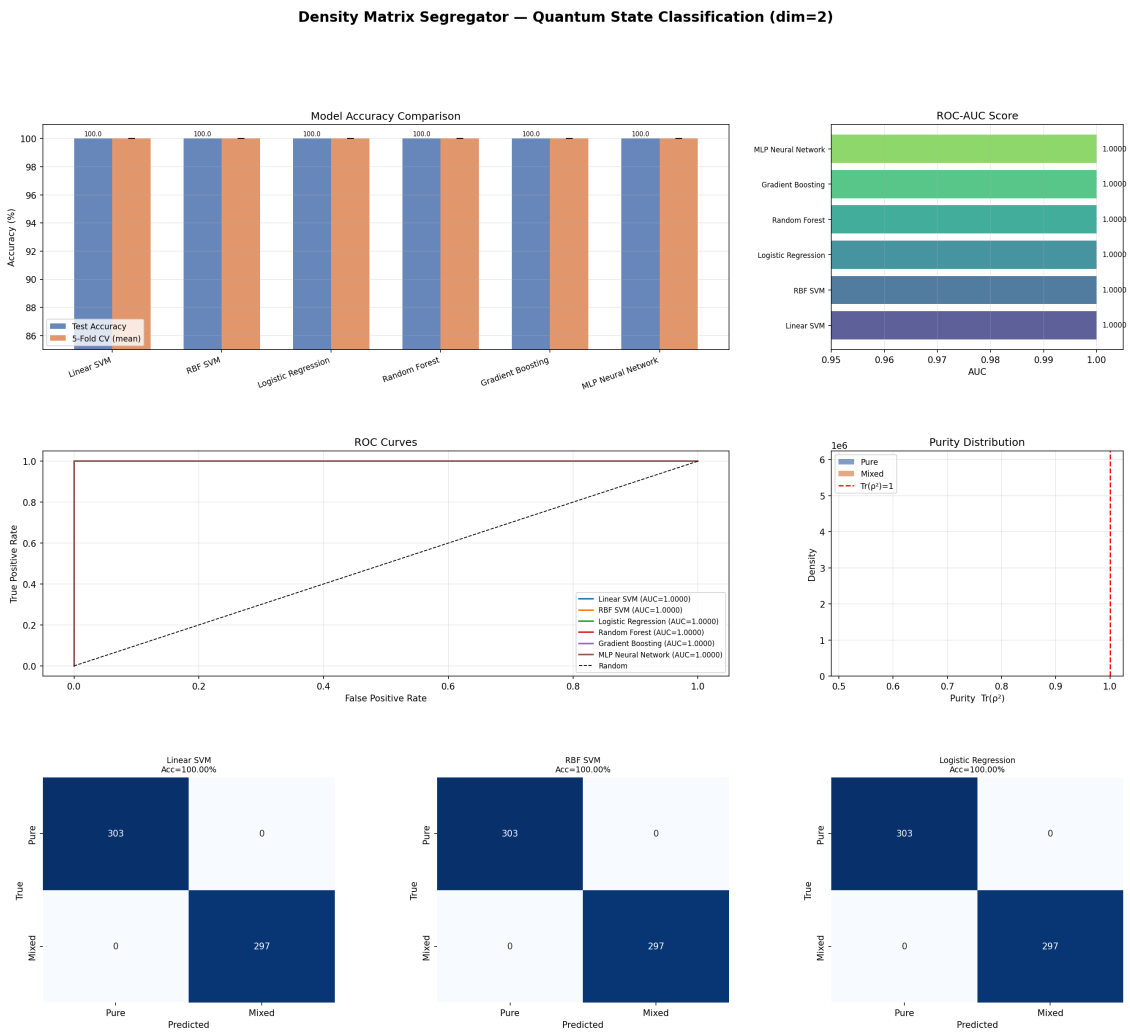


Figure 5.1: Summary of classical machine learning results for density matrix classification.

5.4.4 Quantum Machine Learning Models

Quantum machine learning models were implemented using quantum feature maps, variational quantum classifiers, and quantum kernel methods.

Classical features were encoded into quantum states using rotational gates and entangling operations. Quantum-state overlaps were used to construct kernel matrices for classification.

Variational quantum circuits consisting of parameterized rotation gates and entangling layers were optimized using hybrid quantum-classical training procedures.

5.4.5 Discussion

The results demonstrate that both classical and quantum machine learning models can effectively classify quantum density matrices.

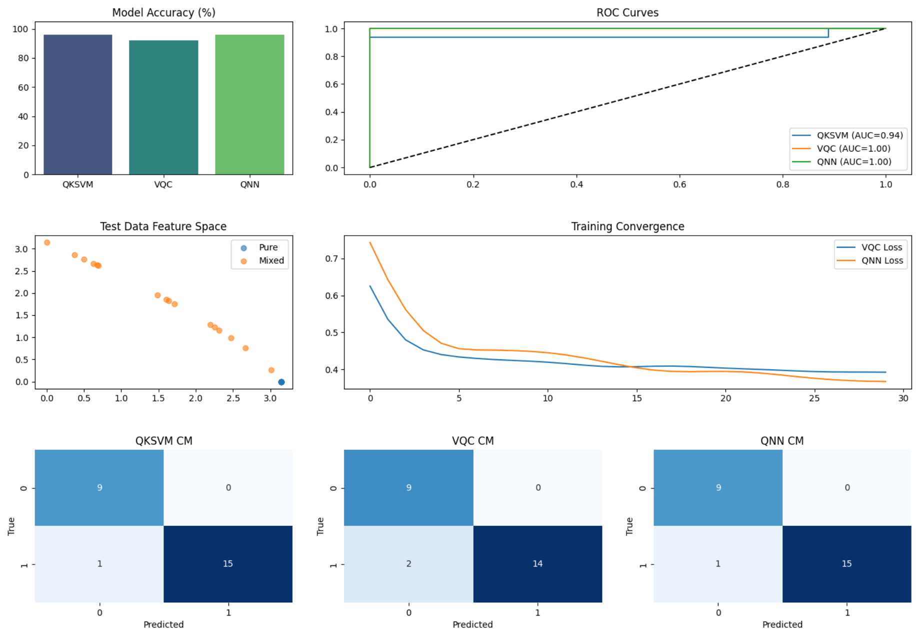


Figure 5.2: Summary of quantum machine learning results for density matrix classification.

For the present dataset sizes, classical machine learning methods already achieve very high accuracies, making quantum approaches somewhat computationally excessive. However, quantum machine learning may become advantageous for larger Hilbert spaces and higher-dimensional quantum systems where classical feature representations become increasingly inefficient.

Furthermore, since the present implementation relied on classical quantities rather than direct quantum-state encoding, the full representational capabilities of quantum machine learning were not completely exploited.

Chapter 6

Machine Learning Methods to understand Sivers Function

6.1 Introduction

In the previous chapter, we looked at various machine learning techniques and their use in segregating density matrices. In this chapter, we shall apply machine learning techniques to extract the Sivers function from data provided in [5].

6.2 Prior analysis in Sivers function

In the papers prior to this work [3][5], Sivers function was extracted by adopting a factorized form for the x and $k_\perp$ dependences, using a Gaussian model for the latter

$$\Delta^N f_{q/p^\uparrow}(x, k_\perp) = 4N_q x^{\alpha_q}(1-x)^{\beta_q}\frac{M_p}{\langle k_\perp^2\rangle_S} k_\perp \frac{e^{-k_\perp^2/\langle k_\perp^2\rangle_S}}{\pi\langle k_\perp^2\rangle}. \tag{6.1}$$

where N_q, α_q and β_q are parameters to be fitted. The SIDIS Sivers asymmetry, defined as [5]

$$A_{UT}^{\sin(\phi_h-\phi_S)} = 2\frac{\int d\phi_S d\phi_h[d\sigma^\uparrow - d\sigma^\downarrow]\sin(\phi_h-\phi_S)}{\int d\phi_S d\phi_h[d\sigma^\uparrow + d\sigma^\downarrow]} = \frac{F_{UT}^{\sin(\phi_h-\phi_S)}}{F_{UU}}, \tag{6.2}$$

can be expressed by the following relations.

$$F_{UT}^{\sin(\phi_h-\phi_S)}(x_B, z_h, P_T, z) = 2z_h \frac{P_T M_p e^{-P_T^2/\langle P_T^2\rangle_S}}{\langle P_T^2\rangle_S^2}\frac{z_h\langle k_\perp^2\rangle_S}{\langle P_T^2\rangle_S}\sum_q e_q^2 N_q x^{\alpha_q}(1-x)^{\beta_q} D_{h/q}(z), \tag{6.3}$$

$$F_{UU}(x, z, P_T, z) = \frac{e^{-P_T^2/\langle P_T^2\rangle}}{\pi\langle P_T^2\rangle}\sum_q e_q^2 f_{q/p}(x) D_{h/q}(z), \tag{6.4}$$

with

$$\langle P_T^2\rangle = \langle p_\perp^2\rangle + z^2\langle k_\perp^2\rangle, \qquad \langle P_T^2\rangle_S = \langle p_\perp^2\rangle + z^2\langle k_\perp^2\rangle_S. \tag{6.5}$$

where,

$$f_{q/p}(x, k_\perp) = f_{q/p}(x)\frac{e^{-k_\perp^2/\langle k_\perp^2\rangle}}{\pi\langle k_\perp^2\rangle}, \tag{6.6}$$

$$D_{h/q}(z, p_\perp) = D_{h/q}(z)\frac{e^{-p_\perp^2/\langle p_\perp^2\rangle}}{\pi\langle p_\perp^2\rangle}, \tag{6.7}$$

where $f_{q/p}(x)$ and $D_{h/q}(z)$ are the unpolarized PDFs and FFs. $\langle k_\perp^2\rangle$ and $\langle p_\perp^2\rangle$ are the widths of the corresponding TMD distributions. The model for the unpolarized TMD PDFs and FFs depend on Q^2 , according to Dokshitzer, Gribov, Lipatov, Altarelli, Parisi (DGLAP) equations [5].

6.3 Machine Learning methods for Sivers function fitting

To model the nonlinear dependence of the SIDIS Sivers asymmetries on the kinematic variables, both classical machine learning (ML) and quantum machine learning (QML) approaches were employed. The input variables consisted primarily of the experimentally relevant kinematic quantities such as Bjorken-x, which were appropriately scaled before training. The data was taken from the paper [5].

6.3.1 Classical Machine Learning Method

The classical ML implementation was based on a Multi-Layer Perceptron (MLP) neural network used for nonlinear regression of the Sivers asymmetries. The neural network learns the mapping between the SIDIS kinematic variables and the corresponding asymmetry observables.

Each neural-network layer performs the transformation

$$h = W\mathbf{x} + b, \tag{6.8}$$

followed by nonlinear activation functions.

The trainable parameters were optimized by minimizing the χ^2 loss function,

$$\chi^2 = \sum_i \frac{(A_i^{\text{exp}} - A_i^{\text{pred}})^2}{\sigma_i^2}, \tag{6.9}$$

where:

- A_i^{exp} denotes the experimental asymmetry,
- A_i^{pred} is the model prediction,
- σ_i represents the experimental uncertainty.

Optimization was performed using the Adam optimizer. Learning-rate scheduling and early stopping techniques were additionally employed to improve convergence stability and reduce overfitting. Monte Carlo replica generation was used to estimate uncertainties in the extracted Sivers functions.

6.3.2 Quantum Machine Learning Method

The QML implementation was based on Variational Quantum Circuits (VQCs) using a hybrid quantum-classical optimization framework.

Classical input data was encoded into quantum states using angle encoding through rotational quantum gates:

$$x \to |\phi(x)\rangle. \tag{6.10}$$

The variational quantum circuit consisted of repeated layers containing:

1. data re-uploading through R_Y rotations,
2. trainable single-qubit rotational gates,
3. entangling CNOT operations.

The quantum state generated by the circuit is represented as

$$|\psi(x,\theta)\rangle = U(\theta)E(x)|0\rangle, \tag{6.11}$$

where:

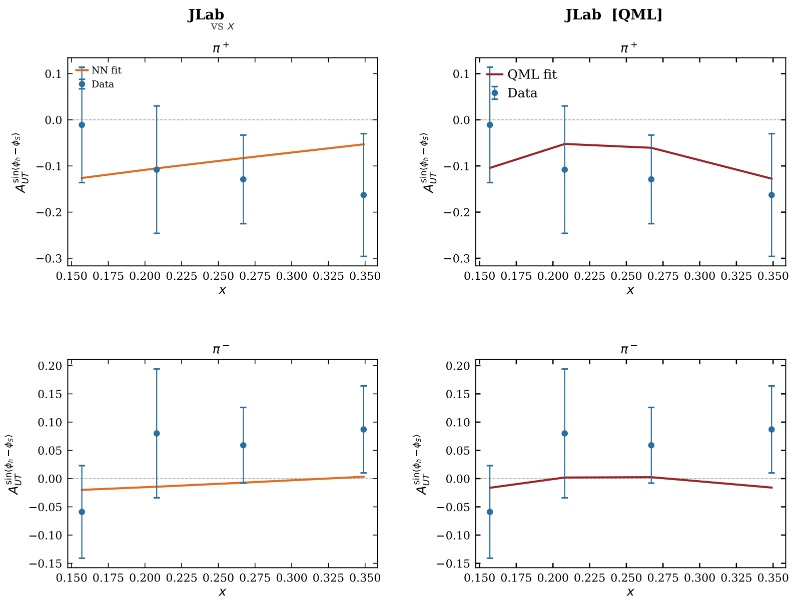
- $E(x)$ is the data-encoding circuit,
- $U(\theta)$ is the trainable variational ansatz,
- θ denotes the trainable quantum parameters.

The circuit outputs were obtained through expectation-value measurements of Pauli observables:

$$\langle Z \rangle = \langle \psi | Z | \psi \rangle. \tag{6.12}$$

Training was performed using hybrid quantum-classical optimization, where the quantum circuit generated predictions and the trainable parameters were updated classically using the Adam optimizer.

Data re-uploading was employed to improve the expressive power of the circuit by repeatedly encoding the input features into multiple variational layers, enabling the model to capture more complex nonlinear structures in the SIDIS asymmetry data.



a) b)

Figure 6.1: Fits prepared using machine learning, data from the JLab experiment (target: ^3He) Here a) is using classical machine learning and b) is using quantum machine learning

6.3.3 Conclusion

In this case, we see that the quantum machine learning wins over classical machine learning. With lesser epochs, it is able to reduce the χ^2 value lesser than the classical case. The summary of the analysis is presented here

Parameters	Fit by [5]	Classical fit	Classical Machine Learning	Quantum Machine Learning
χ^2	212.8	388.3	368.0	364.9
epochs	-	-	3000	450
N_u	0.4 ± 0.09	0.3626 ± 0429	0.2516 ± 0.0034	0.2210 ± 0.0067
β_u	5.430 ± 1.590	5.3681 ± 0.8931	3.0089 ± 0.1450	2.4202 ± 0.2800
N_d	-0.630 ± 0.230	-0.9069 ± 0.2958	-0.9346 ± 0.0023	-0.6354 ± 0.0077
β_d	$+0.300 \pm 0.150$	$+0.2969 \pm 0.0758$	$+0.3079 \pm 0$	$+0.3000 \pm 0$

Results of ML and QML fitting of the given data

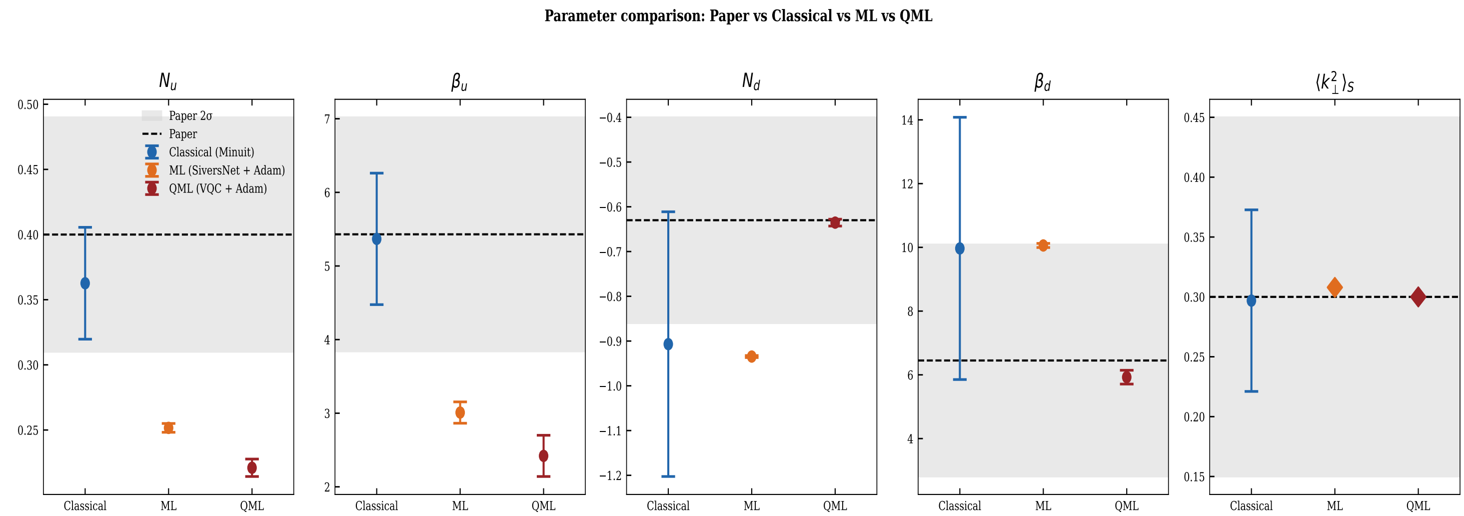


Figure 6.2: Comparison of various parameters obtained using classical fit, machine learning and quantum machine learning with one obtained in the paper [5].

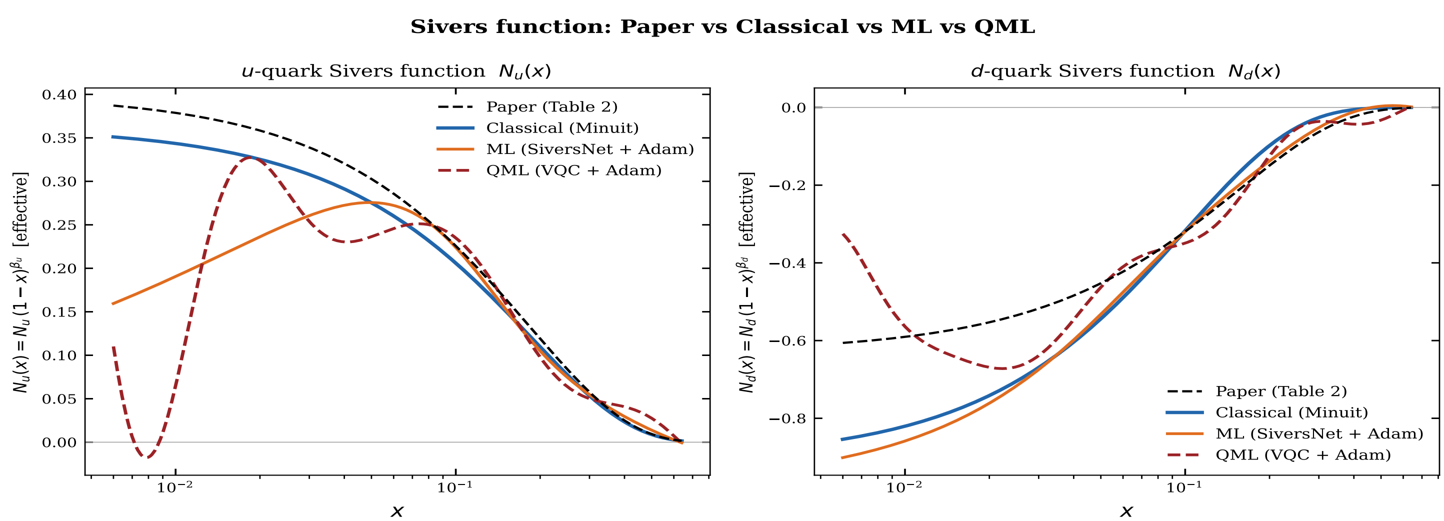


Figure 6.3: Comparison of the Sivers function obtained using classical fit, machine learning and quantum machine learning with one obtained in [5]

Chapter 7

Conclusions and Future work

7.1 Conclusions

This project was an attempt to look at certain aspects of the spin structure of the proton. First, we studied elastic, inelastic and deep inelastic scattering and derived the cross section formulae using structure functions. Then we extended the idea to polarized deep inelastic scattering where we included the effect of spin with the deep inelastic scattering case. Then we went into semi inclusive deep inelastic scattering where we found out the structure functions. From there we extracted information about the Sivers function and we also had discussed a method to find out the function experimentally using transverse single spin assymmetries.

The second part of the project dealt with machine learning, both classical and quantum. We explained the process behind each technique and tried it on a trial problem of segregating density matrices. After that, we tried the same on trying to fit data to obtain the Sivers function. We also found out that with the technique we used, even though quantum machine learning didn't give a huge advantage in terms of segregating density matrices, it gave a good advantage when it comes to fitting the Sivers function.

With this project, we conclude that the structure of the proton in particular, the Sivers function can be reasonably understood using machine learning methods.

7.2 Future work

This project has opened a way to use quantum machine learning to fit a function as complex as the Sivers function. Efforts can be made to optimize the neural network algorithm better. A possible direction is to try use these machine learning methods to segregate Sivers and Collins functions. We can also look into analysis of more such functions which do not have an initial fitting function, especially in particle physics.

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